

Client Setup

The frontend was scaffolded using Vite for extremely fast Hot Module Replacement (HMR) and modern build tooling.

Setup Commands

```
cd client
npm create vite@latest . -- --template react
npm install
```

Key Dependencies Installed

- `react-router-dom` : For handling multi-page navigation (Login, Dashboard, Ride Tracking).
- `react-leaflet` & `leaflet` : For rendering the interactive map and markers.
- `socket.io-client` : For connecting to the backend WebSocket server.
- `axios` : For making HTTP requests to the REST API.
- `react-icons` : For SVG iconography.